

PHARMASENTINEL AI

Detect signals. Find gaps. Act earlier.

Drug Safety Signal Detection + ICH M4 CTD Submission Readiness



THE PROBLEM

Massive adverse-event data hides emerging safety signals

Millions of report associations

Duplicate and inconsistent data

Thousands of drug-event combinations

Large adverse reaction term volumes

Detect emerging safety signals early – before they become difficult to investigate.

Complex CTD dossiers make regulatory gaps easy to miss

Structured documentation across CTD modules

Missing or incomplete sections cause delays and risk

Know what is missing before submission.

ONE PLATFORM. TWO CRITICAL PROBLEMS.

ONE PLATFORM. TWO WORKFLOWS.

SIGNAL DETECTION

FAERS/
AEMS Data → Normalize → Analyze → PRR Investigator → Prioritize →



PHARMASENTINEL

AI

SUBMISSION READINESS

CTD Outline → Check Against ICH M4 → Measure Completeness → Identify Gaps → Prioritize



From safety intelligence to regulatory readiness —
in one decision platform.

DETECT

Search signals...

All Drugs

All Events

All Confidence

Reset

Drug	Adverse Event	PRR	X	Confidence	Priority
Drug A	Event X	4.82	31.4	High	Critical
Drug B	Event Y	2.91	18.7	High	High
Drug C	Event Z	1.87	8.4	Moderate	Medium



SELECTED SIGNAL DETAIL

Drug AEvent X

2x2 evidence breakdown

A
Drug + Event

B
Drug + Other Events

C
Other Drugs + Event

D
Other Drugs + Other Events

PRR =

A / (A + C)

B / (B + D)

Pearson

2

(AD - BJ)

(A + B + C + D)

(A + B)(C + D)(A + C)(B + D)

A high level signal is a safety concern requiring investigation, but not necessarily urgent.

REVIEW

System compares dossier outline with ICH M4 requirements to highlight gaps.



GAP REPORT

- CRITICAL (3)** Missing required sections
- HIGH (5)** Incomplete supporting information
- MEDIUM (8)** Documentation improvement required

M1 ADMINISTRATIVE 100% COMPLETE ✓	M2 SUMMARIES 92% COMPLETE ✓	M3 QUALITY 81% ATTENTION ⚠	M4 NONCLINICAL 90% COMPLETE ✓	M5 CLINICAL 84% ATTENTION ⚠
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Module	M3 – Quality
Section	M3.2.P.5 Control of Drug Product
Status	MISSING
Priority	CRITICAL
Recommended action	Review and complete required section

Readiness score is an application-level completeness indicator, not an official regulatory approval prediction.

HOW IT WORKS

DATA

raw inputs

PROCESSING

clean and structure

ANALYTICS

detect and quantify

APPLICATION

deliver insights

AI / WORKFLOW

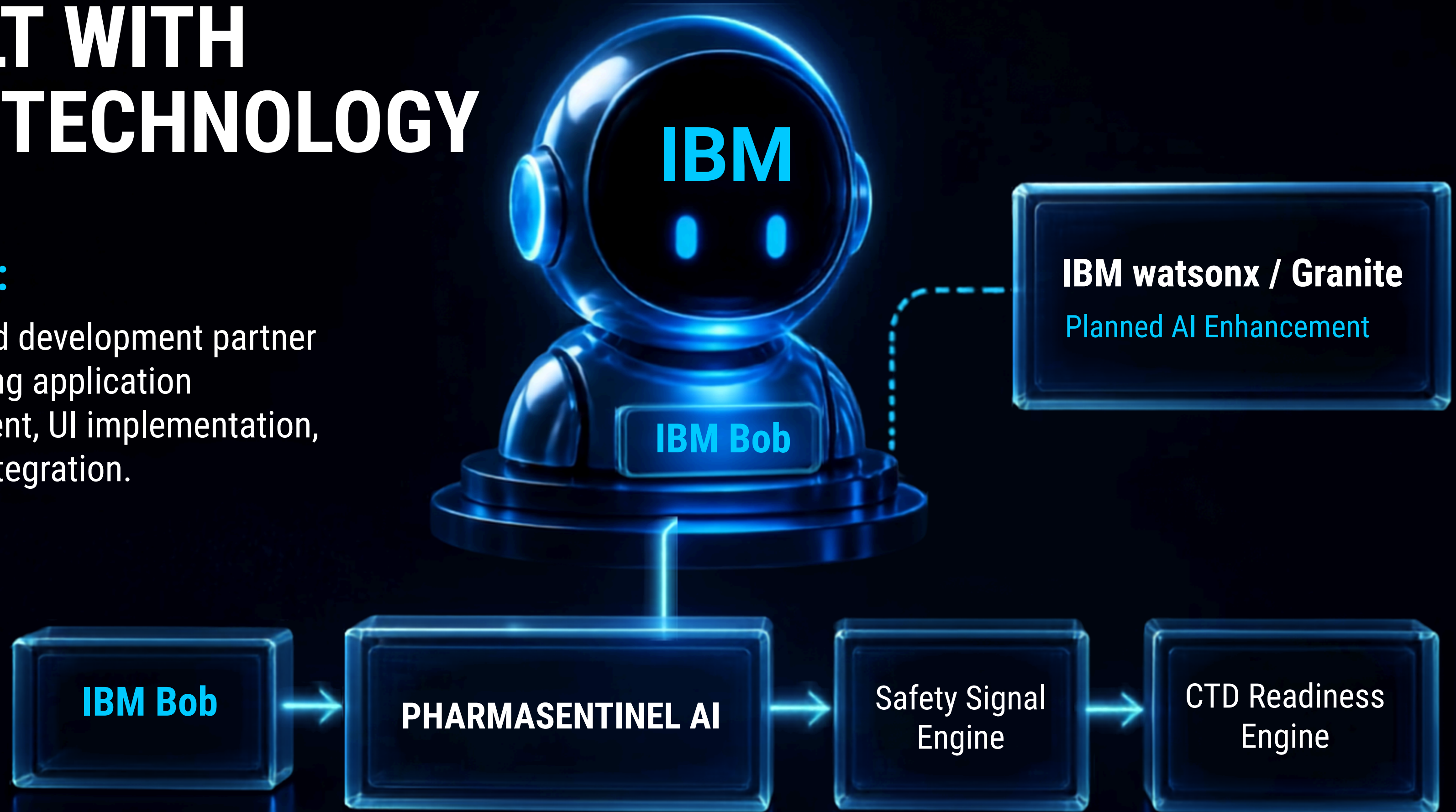
orchestrate and act



BUILT WITH IBM TECHNOLOGY

IBM Bob:

AI-powered development partner
accelerating application
development, UI implementation,
and API integration.



IMPACT

397K+

reports

4,841

active
ingredients

12,389

adverse
reaction terms

633K+

drug-event
pairs

10M+

associations

34/34

automated
tests passing

TODAY

automated signal screening
CTD completeness checking
prioritized investigations and gaps

BEYOND THE HACKATHON

continuous pharmacovigilance
smarter regulatory workflows
earlier safety signal identification
transparent submission preparation

PHARMASENTINEL AI

DETECT EARLIER.

REVIEW SMARTER.

SUBMIT WITH CONFIDENCE.



AI-powered pharmaceutical safety & regulatory intelligence